Biperiodization: the EMF method

Benjamin Ménétrier

Last update: February 20, 2025

1 Problem definition

The biperiodization process is needed to make a non-periodic 2D field periodic, in order to later apply discrete Fourier transforms. For this, an extension zone is defined on two sides of the original domain (called the "physical zone"), which has to be filled with adequate values to make the whole field periodic.

The work of Blažica *et al.* (2015) reviews existing methods, and can be quickly summarized as follows:

- The extension methods of HIRLAM and ALADIN models are based on spline (ALADIN) or trigonometric (HIRLAM) functions, which are applied in both directions successively. They lead to wild overshoots in the extension zone and have a strong impact on the spectra.
- The detrending approach requires a modification of the physical values.
- The Boyd method is very good but requires data from outside the physical zone (e.g. from a large scale coupler).
- The discrete cosine transform method produces accurate spectra but increases the total domain size by a factor 4.

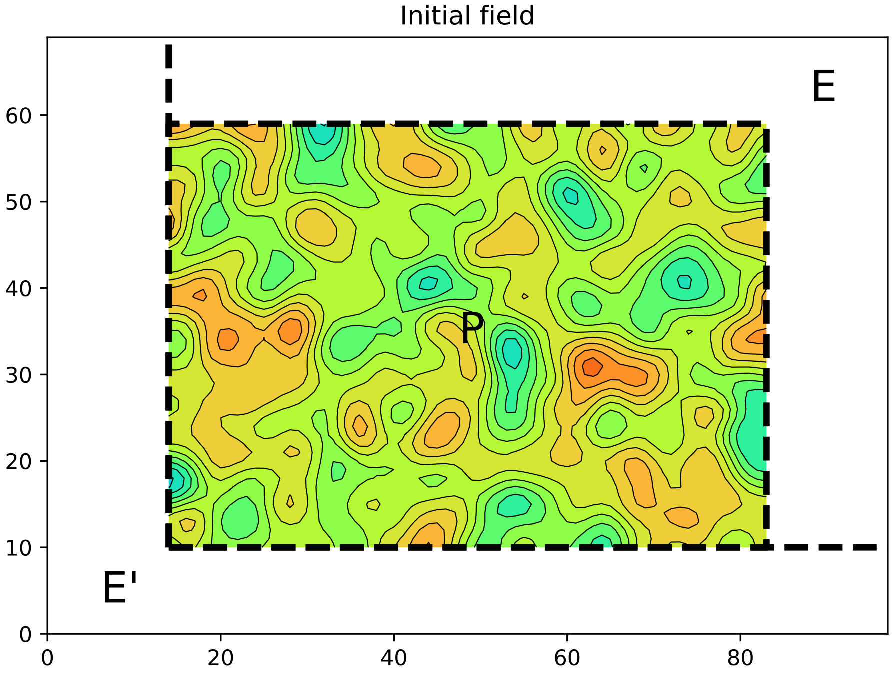
2 The EMF method

The goal of this note is to design a method that:

- produces smooth extension values without overshoots,
- does not require a modification of the physical values,
- can be computed from the physical values only,
- does not increase the total domain size too much.

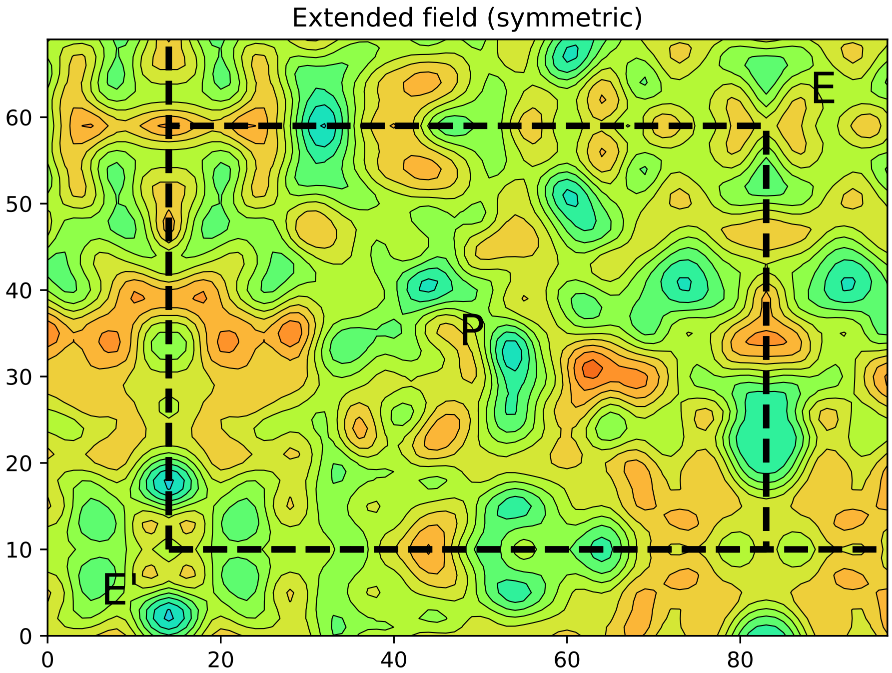
This new (?) method is called "EMF" for "Extension, Mixing, Folding".

We start with an initial random smooth field defined in the physical zone (P). The goal is to define values in the extension zone (E), and to make this process easier we also fill the opposite extension zone (E').



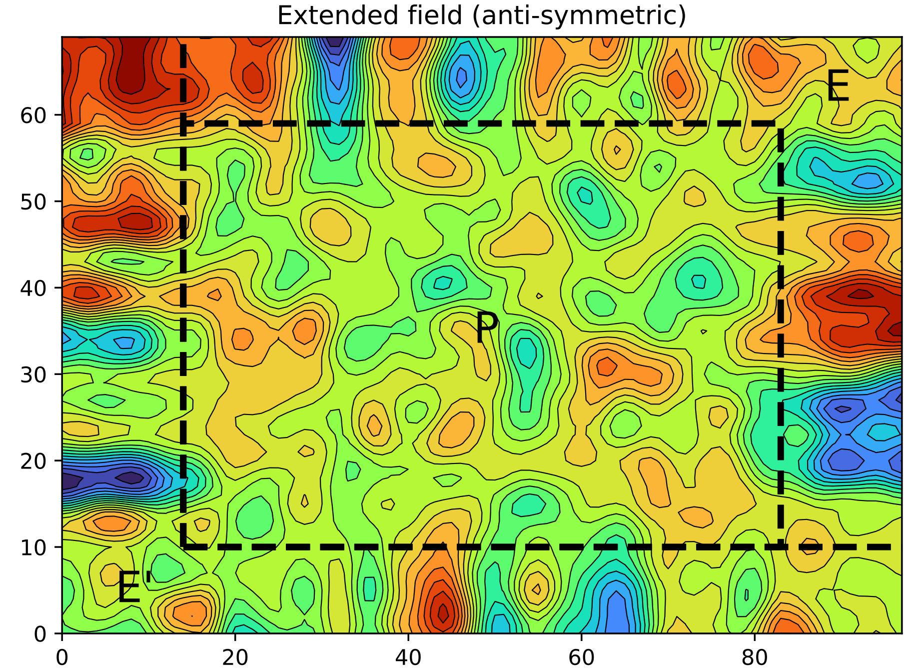
2.1 Extension

As in the discrete cosine transform method, it is possible to extend the initial field using a symmetry operation with respect to the (P) boundary:



The values in (E) and (E') have the same amplitude as in (P), which is good, but the field derivative is discontinuous at the (P) boundary, which is not ideal.

Another possible way to extend the field is to use an anti-symmetric operation with respect to the (P) boundary:



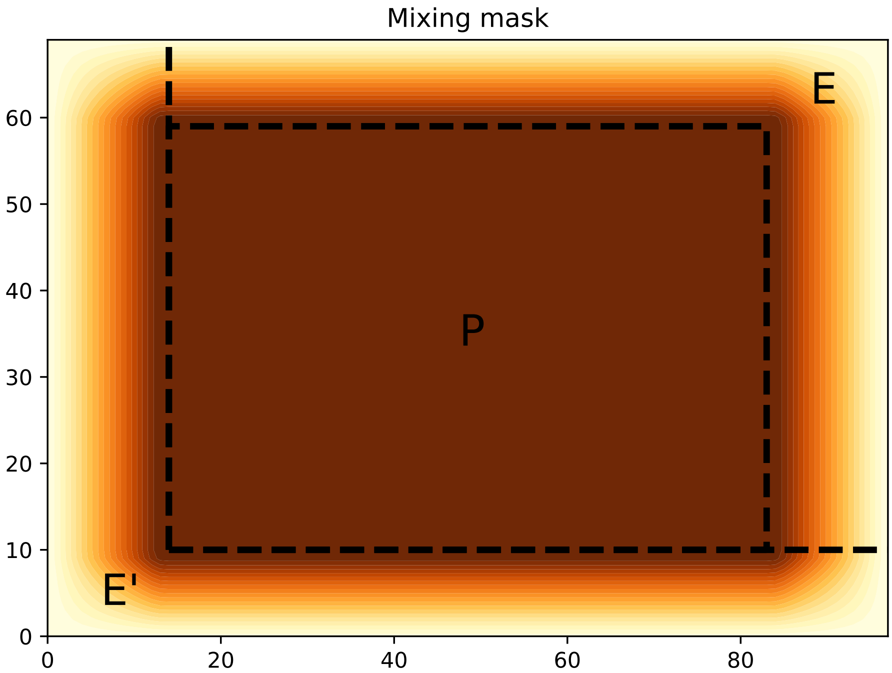
In this case, the first two derivatives of the field are continuous at the (P) boundary, which is good, but the field shows significant overshoots in (E) and (E') further away from this boundary, which is not desired.

2.2 Mixing

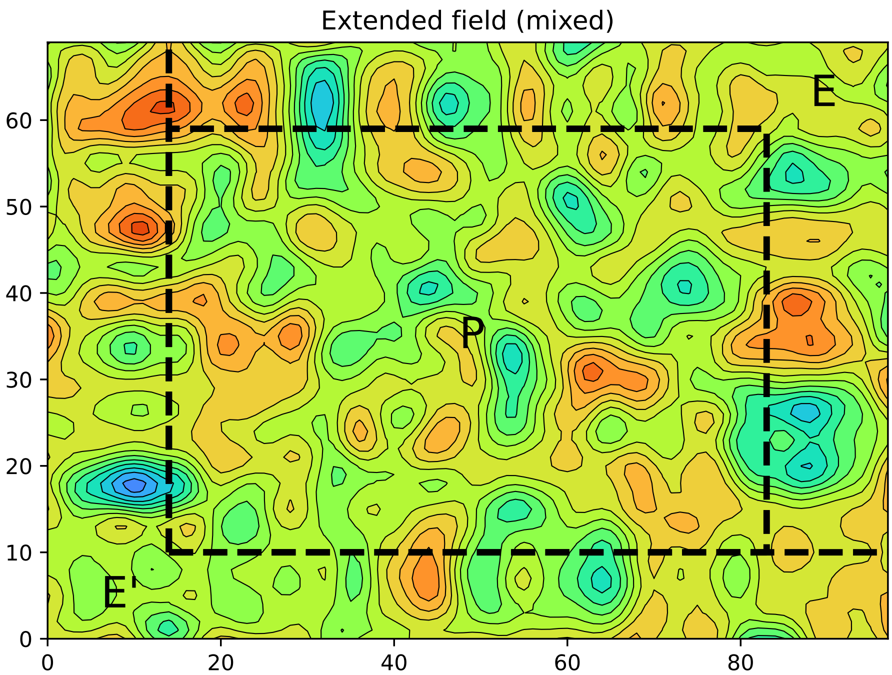
To attenuate the respective drawbacks of the two previous extension methods, a simple linear combination might be sufficient. The weights of this combination should be:

- higher on the first field (symmetric) away from the boundary, to avoid overshoots,
- higher on the second field anti-symmetric) at the boundary, to keep continuous derivatives.

A mixing mask using Boyd (2005) formula for the ramp can be defined:



This mask is applied to the anti-symmetric extension, and its complementary to the symmetric extension. The mixed field resulting from this linear combination is:



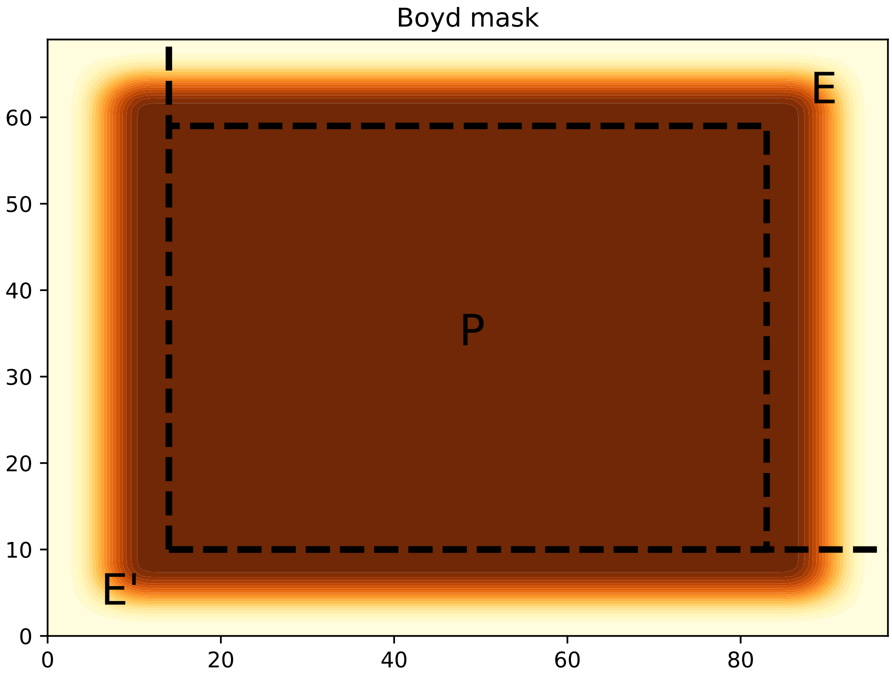
The continuity of the derivatives is preserved at the (P) boundary, while overshoots are kept at a managable amplitude.

2.3 Folding

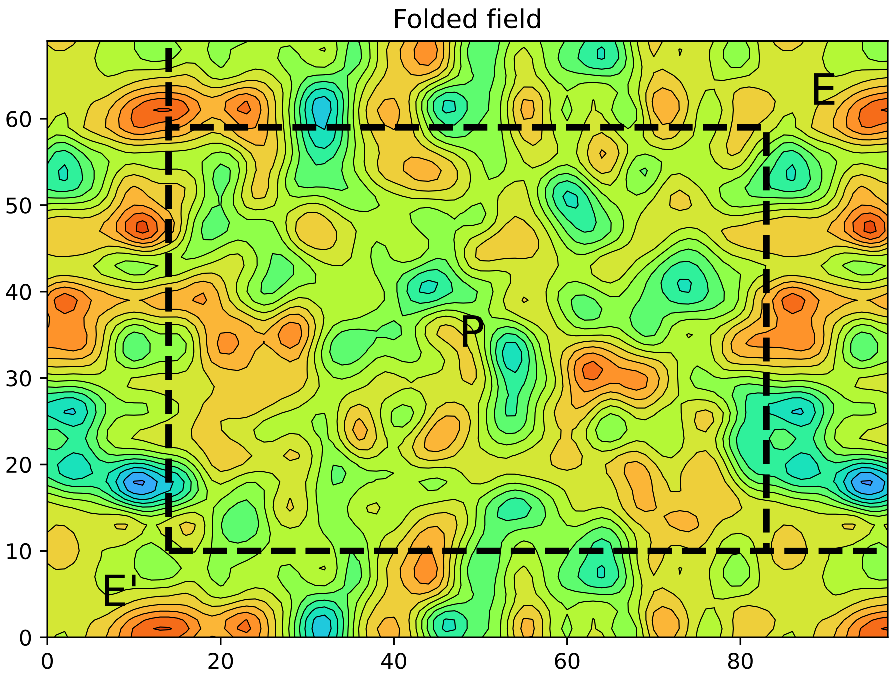
In the final step, we apply the Boyd (2005) folding method on this extended field, following two steps:

- applying the Boyd's mask on the extended field,
- folding the domain to make the (P+E) zone biperiodic.

The Boyd's mask look like:



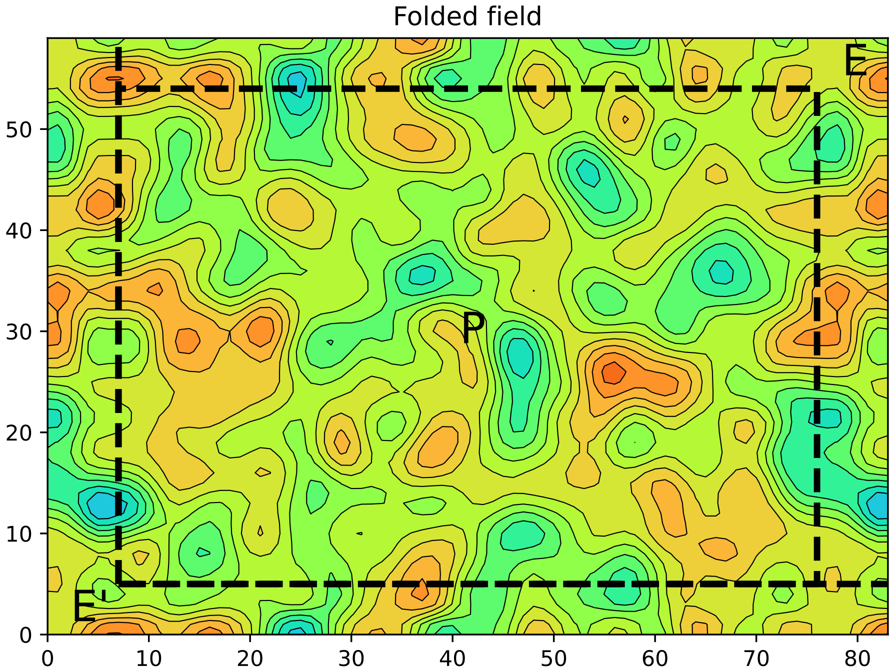
and the final result is:



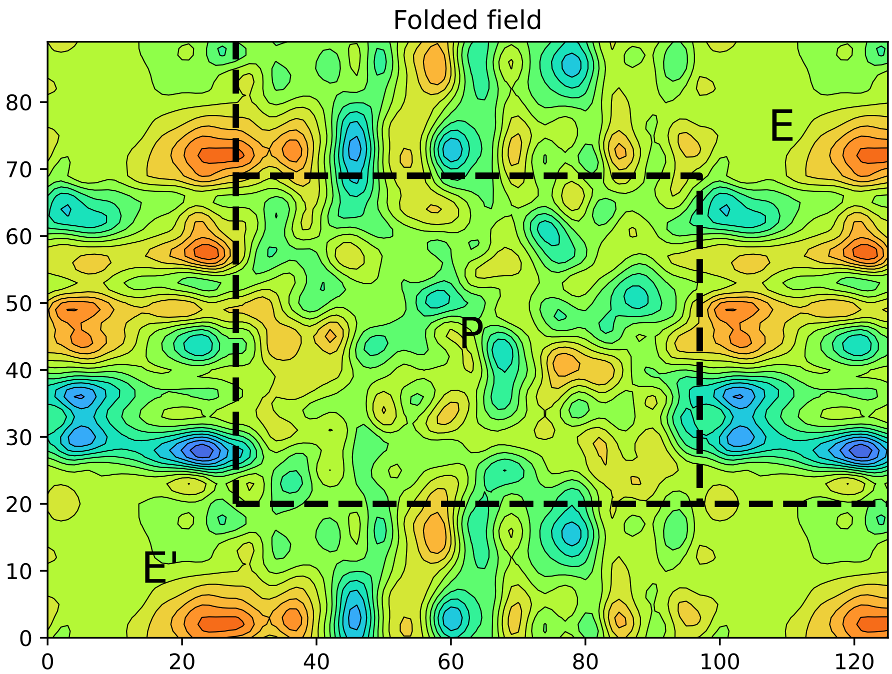
We can check that the (P+E) zone is now biperiodic, with continuous derivatives at the (P) boundary and limited overshoots in (E). The field in (P+E) has been built from the values of (P) only, without modifying them.

3 Impact of the extension zone size

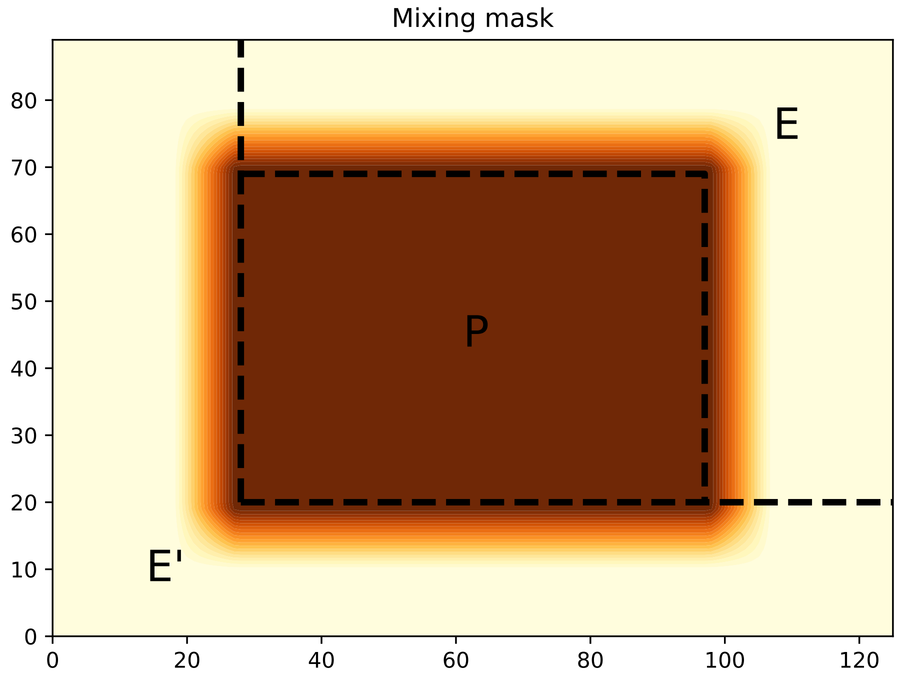
With an extension zone twice narrower, the EMF method works fine and the final result is:



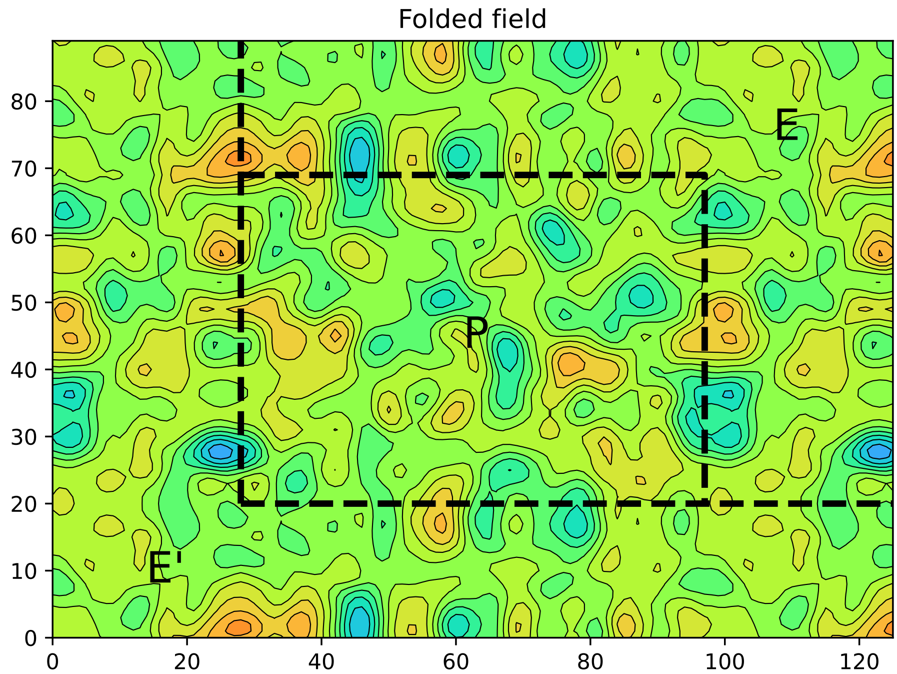
With and extension zone twice broader, larger overshoots produced by the anti-symmetric extension are expected:



However, it is easy to adjust the mixing mask with a transition that is not in the middle of (E) and (E') but closer to the (P) boundary instead:



leading to a final result with limited overshoots:



References

Blažica V, Gustafsson N, Žagar N. 2015. The impact of periodization methods on the kinetic energy spectra for limited-area numerical weather prediction models. *Geoscientific Model Development* **8**(1): 87–97.

Boyd JP. 2005. Limited-area fourier spectral models and data analysis schemes: Windows, Fourier extension, Davies relaxation, and all that. *Monthly Weather Review* **133**(7): 2030–2042.